



Year 11 Mathematics Extension 1
Assessment Task 1
Term 2, Week 6, 2023

Name:

Teacher:

Monday 29th May 2023

Setter: CHS

General Instructions

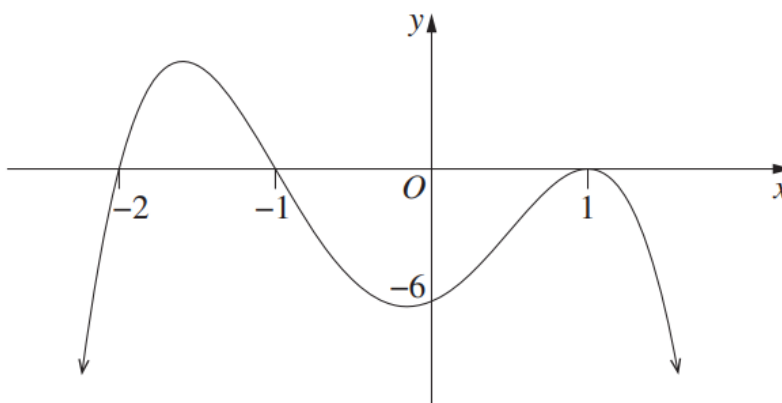
- **Working time – 45 minutes + 2 minutes reading time.**
 - Answer all questions on the lined paper provided using black or blue pen.
 - NESA approved calculators may be used.
 - Show all relevant mathematical reasoning and/or calculations.
 - **Total marks – 32**
 - Attempt Questions 1–5
-

Question 1 (7 marks)

Marks

- (a) The diagram shows the graph of $y = a(x+b)(x+c)(x+d)^2$.

1



What are the possible values of a , b , c and d ?

- (A) $a = -6$, $b = -2$, $c = -1$, $d = 1$
- (B) $a = -6$, $b = 2$, $c = 1$, $d = -1$
- (C) $a = -3$, $b = -2$, $c = -1$, $d = 1$
- (D) $a = -3$, $b = 2$, $c = 1$, $d = -1$

Question 1 continues on the next page

- (b) The roots of $x^3 - 2x^2 + 5x + 1 = 0$ are α, β and γ . Find the value of:
- (i) $5\alpha + 5\beta + 5\gamma$. **1**
- (ii) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$. **1**
- (iii) $\alpha^2 + \beta^2 + \gamma^2$. **2**
- (c) Let $P(x) = x^3 - ax^2 + x$ be a polynomial where a is a real number. When $P(x)$ is divided by $(x-3)$ the remainder is 12. Find the remainder when $P(x)$ is divided by $(x+1)$. **2**

End of Question 1

(a) Which inequality has the same solution as $|x+2|+|x+3|=5$?

1

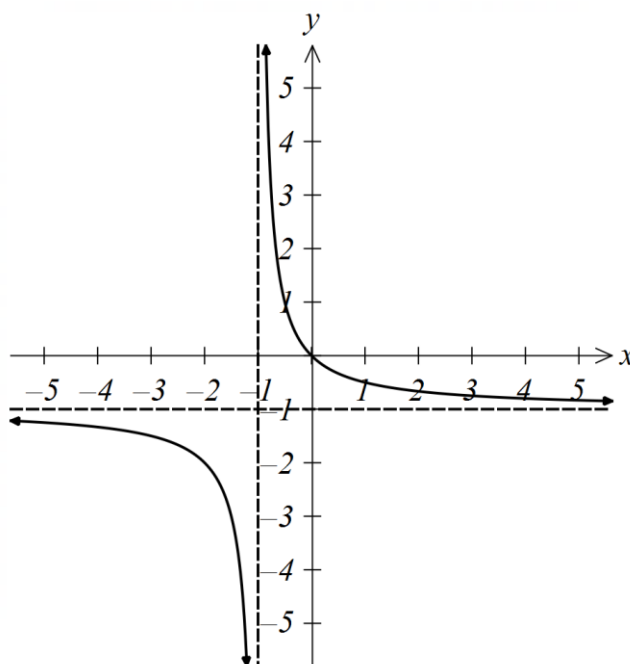
(A) $\frac{5}{3-x} \geq 1$

(B) $\frac{1}{x-3} - \frac{1}{x+2} \leq 0$

(C) $x^2 - x - 6 \leq 0$

(D) $|2x-1| \geq 5$

(b) The diagram below is a sketch of the graph of the function $f(x) = -\frac{x}{x+1}$.



(i) Sketch the graph $y = (f(x))^2$, showing all the asymptotes and intercepts.

2

(ii) Sketch the graph $y = x + f(x)$, showing all asymptotes and intercepts.

2

(iii) Solve the equation $(f(x))^2 = f(x)$.

2

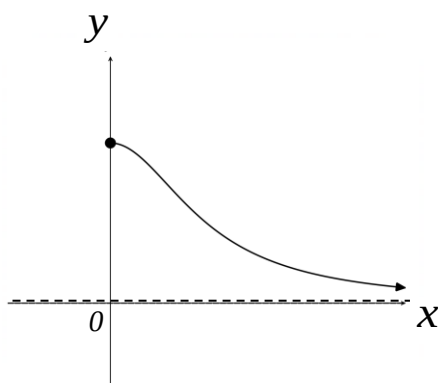
End of Question 2

(a) Solve $\frac{x^2 + 5}{x} > 6$.

2

(b)

2

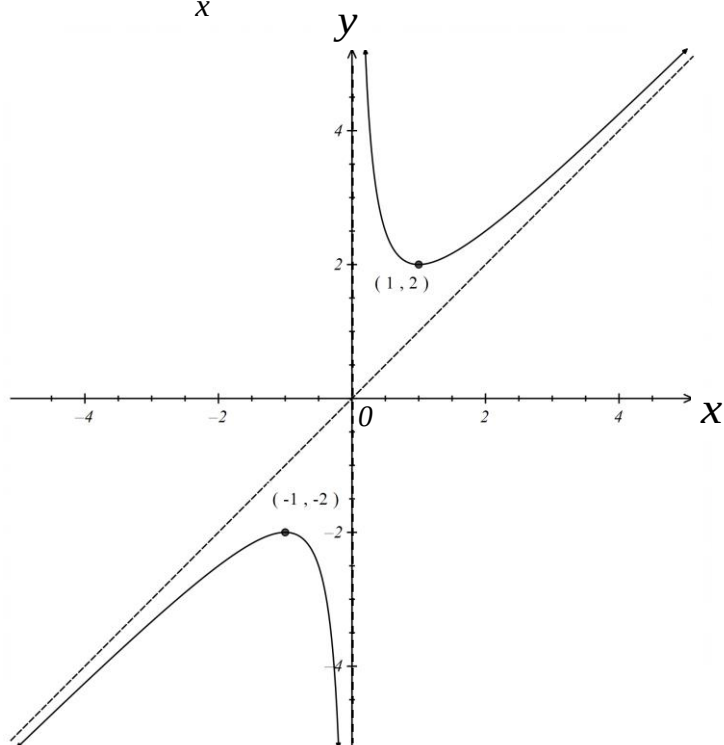


The diagram above shows the function $f(x) = \frac{1}{x^2 + 1}$ for $x \geq 0$.

State the domain and range of the inverse function $f^{-1}(x)$.

(c) The graph of the function $f(x) = \frac{x^2 + 1}{x}$ is shown.

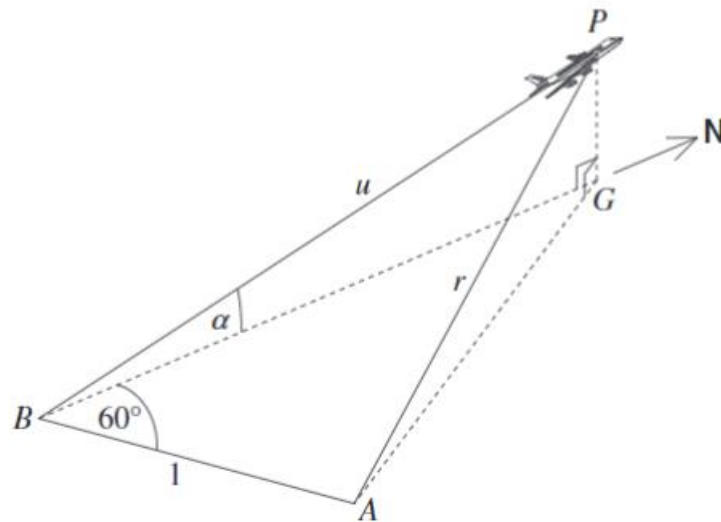
2



Sketch the graph of $y = \frac{1}{\sqrt{f(x)}}$, showing all important features including turning point(s), intercept(s) and asymptote(s). Your sketch is required to be at least a third of the page.

End of Question 3

- (a) A monic polynomial $P(x)$ of degree 4 has one zero equal to 2 and a double zero equal to -1 . 3
If the polynomial passes through the point $(0,6)$, find the equation of $P(x)$.
- (b) A plane P takes off from a point B . It flies due north at a constant angle α to the horizontal. 3
An observer is located at A , 1 km from B , at a bearing 060° from B . Let u km be the distance from B to the plane and let r km be the distance from the observer to the plane. The point G is on the ground directly below the plane.



Show that $r = \sqrt{1 + u^2 - u \cos \alpha}$.

End of Question 4

Question 5 (6 marks) (Start a new page)**Marks**

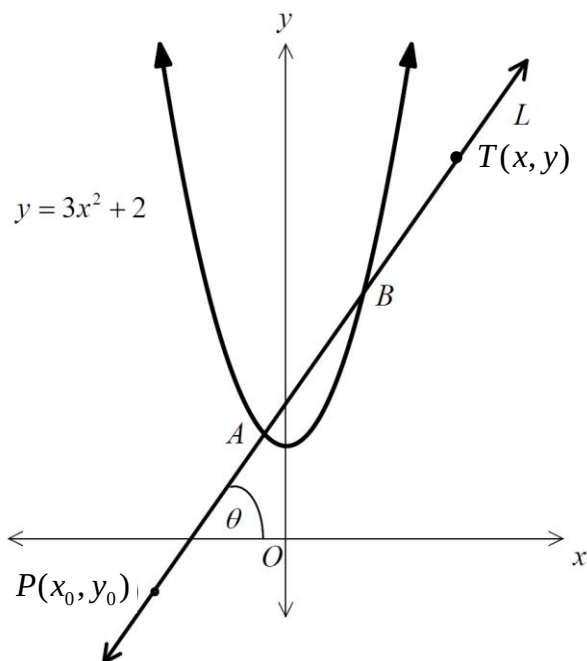
Let L be the straight line passing through a fixed point $P(x_0, y_0)$ with an angle of inclination θ to the positive x -axis. The line L is defined parametrically by:

$$x = x_0 + t \cos \theta$$

$$y = y_0 + t \sin \theta$$

Where t is real and $0 \leq \theta \leq 180^\circ$.

In the diagram below the line L cuts the parabola $y = 3x^2 + 2$ at points A and B , where P is a fixed point taking the coordinates $\left(-1, -\frac{1}{3}\right)$.



(i) $|PA| = t_1$ and $|PB| = t_2$. Show that for any point T on the line L that $|PT| = |t|$. **1**

(ii) Show that t_1 and t_2 are the roots of the equation: **2**

$$9t^2 \cos^2 \theta - 3t(\sin \theta + 6 \cos \theta) + 16 = 0.$$

(iii) Given that $|AB|^2 = \frac{\sin^2 \theta + 12 \sin \theta \cos \theta - 28 \cos^2 \theta}{9 \cos^4 \theta}$ (Do **NOT** prove this). **2**

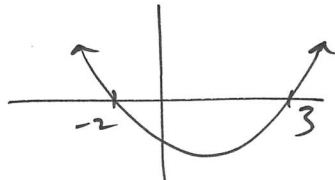
and L is a tangent to the parabola $y = 3x^2 + 2$, show that the two possible gradients of L are 2 and -14 .

(iv) If Z is the point of contact of the tangent L with the parabola $y = 3x^2 + 2$ and L has a gradient of 2, show that $|PZ| = \frac{4\sqrt{5}}{3}$. **1**

End of Assessment Task



2023 Year 11 Mathematics Extension 1 Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>QUESTION 1</u> a) single root @ $x = -2, -1$ double root @ $x = 1$. possible values of 'b' & 'c' $b = 2$ $c = 1$ $d = -1$ $y = a(x+2)(x+1)(x-1)^2$ let $x=0, y=-6$ (y int). $-6 = a(2)(1)(-1)^2$ <u>$a = -3$</u> $\therefore$ Answer: D b) i) $5(x+B+r)$ $-\frac{b}{a} = 2 \Rightarrow 5(2)$ $= 10 \checkmark$ ii) $x^2 B r + x B^2 r + x B r^2$ $x B r = -\frac{d}{a}$ $= -1$ $x B r (x+B+r) = (-1)(2)$ $= -2 \checkmark$ iii) $x^2 + B^2 + r^2$ $(x+B+r)^2 - 2(xB + Br + x r)$ $(2)^2 - 2(5)$ $4 - 10$ $-6 \checkmark$		c) $P(x) = x^3 - ax^2 + x$ $P(3) = 12$ $12 = (3)^3 - a(3)^2 + 3$ $12 = 30 - 9a$ <u>$a = 2$</u> $\checkmark$ d) $P(-1) = (-1)^3 - 2(-1)^2 + (-1)$ $= -1 - 2 - 1$ $= -4 \checkmark$ <u>QUESTION 2</u> a) if $x < -2, x-3 > 5$ $\therefore$ no solution for $x < -2$ if $x > 3, x+2 > 5$ no solution for $x > 3$ if $-2 \leq x \leq 3$ $x+2 - (x-3) = 5$ $x+2 - x+3 = 5$ $5 = 5$ looking at $x^2 - x - 6 \leq 0$ $(x-3)(x+2) \leq 0$  <u>Answer: C</u>	$\textcircled{D}$ solving 'a' $\textcircled{D}$ answer
	$\textcircled{D}$ answer		
	$\textcircled{D}$ answer		

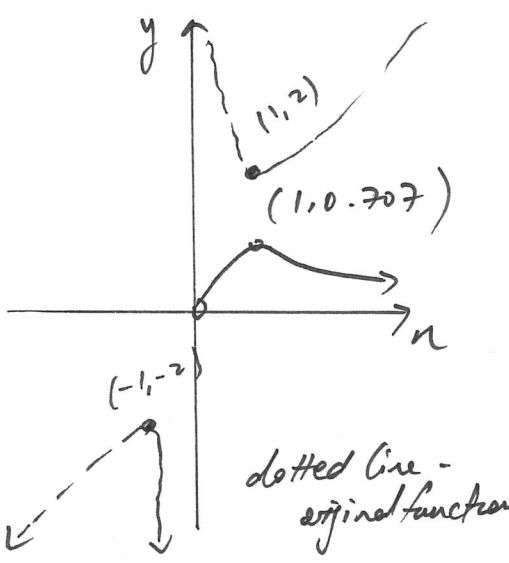
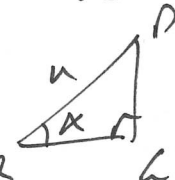
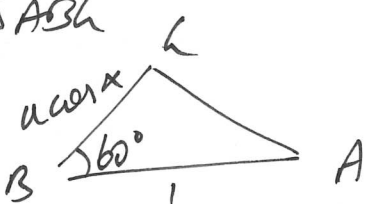
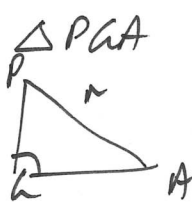


2023 Year 11 Mathematics Extension 1 Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
b) i) ii) iii) $\left(\frac{-x}{x+1}\right)^2 = -\frac{x}{x+1}$ $\frac{x^2}{(x+1)^2} = -\frac{x}{x+1}$ $x^2(x+1) = -x(x+1)^2$ $x^3 + x^2 = -x^3 - 2x^2 - x$ $2x^3 + 3x^2 + x = 0$ $x(2x^2 + 3x + 1) = 0$ $x(x+1)(2x+1) = 0$	① correct shape ① correct asymptotes -1 for any incorrect/missing features. ① correct shape ① key points & asymptotes	$x=0$ $y=0$ $x \neq -1$ $x = -\frac{1}{2}$ $y=1$ QUESTION 3 a) $\frac{x^2+5}{x} > 6$ $x(x^2+5) > 6x^2$ $x^3 - 6x^2 + 5x > 0$ $x(x^2 - 6x + 5) > 0$ $x(x-1)(x-5) > 0$ $\therefore 0 < x < 1$ or $x > 5$ b) D: $0 < x \leq 1$ ✓ R: $y \geq 0$ ✓ c) at $x=1 \Rightarrow f(1)=2$ (1,2) @ $y = \frac{1}{\sqrt{f(x)}} \Rightarrow \frac{1}{\sqrt{2}} = (1, 0.707)$ at $x=-1 \Rightarrow f(-1)$ no soln at $x \rightarrow 0 \Rightarrow$ no soln $x > 0$ for graph to exist as $x \rightarrow \infty, y \rightarrow 0^+$	① soln must include $x \neq -1$. ① working ① answer ① each correct
	① factorise		

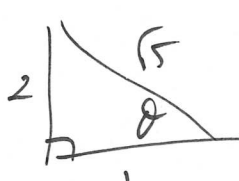


2023 Year 11 Mathematics Extension 1 Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
 QUESTION 4 a) one zero @ $x = 2$ double zero @ $x = -1$ $P(x) = (x-2)(x+1)^2(x-\alpha)$ passes thru $(0, 6)$ $6 = (0-2)(0+1)^2(0-\alpha)$ $6 = (-2)(1)(-\alpha)$ $6 = 2\alpha$ $\alpha = 3$ $\therefore P(x) = (x-2)(x+1)^2(x-3)$	① open circle @ $x=0$ ① correct shape & approaching $y \rightarrow 0^+$	b) $\triangle PBH : \sin \alpha = \frac{PH}{u}$ $PH = u \sin \alpha$ $\cos \alpha = \frac{BH}{u}$ $BH = u \cos \alpha$  $\triangle ABH$  $AH^2 = BH^2 + AB^2 - 2(BH)(AB) \cos \alpha$ $AH^2 = (u \cos \alpha)^2 + 1 - 2(u \cos \alpha)(1) \times \frac{1}{2}$ $= u^2 \cos^2 \alpha + 1 - u \cos \alpha$ $\triangle PGA$  $PA^2 = PG^2 + AG^2$ $r^2 = (u \sin \alpha)^2 + u^2 \cos^2 \alpha + 1 - u \cos \alpha$ $= u^2 \sin^2 \alpha + u^2 \cos^2 \alpha + 1 - u \cos \alpha$ $= u^2 (\sin^2 \alpha + \cos^2 \alpha) + 1 - u \cos \alpha$ $= u^2 + 1 - u \cos \alpha$ $r = \sqrt{1 + u^2 - u \cos \alpha}$	① cosine rule AH ① using $\triangle PGA$ for PA^2 ① answer



2023 Year 11 Mathematics Extension 1 Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>QUESTION 5</u> (i) $P(x_0, y_0)$ $T(x, y)$ $PT = \sqrt{(x-x_0)^2 + (y-y_0)^2}$ $PT = \sqrt{t^2 (\cos^2 \theta + \sin^2 \theta)}$ $= \sqrt{t^2}$ $PT = t $ ii) let $x_0 = -1, y_0 = \frac{1}{3}$ $y = 3x^2 + 2 \Rightarrow \text{sub } x = x_0 + t \cos \theta$ $y = y_0 + t \sin \theta$ $y_0 + t \sin \theta = 3(x_0 + t \cos \theta)^2 + 2$ sub x_0, y_0 $\frac{1}{3} + t \sin \theta = 3(-1 + t \cos \theta)^2 + 2$ $\times 3$ $-1 + 3t \sin \theta = 9 - 18t \cos \theta + 9t^2 \cos^2 \theta + 6$ $0 = 9t^2 \cos^2 \theta - 18t \cos \theta - 3t \sin \theta + 16$ $0 = 9t^2 \cos^2 \theta - 3t(\sin \theta + 6 \cos \theta) + 16$ answer -	① correct use of parametric ① correct sub x_0, y_0	iii) gradient $m_L = \frac{y-y_0}{x-x_0} = \frac{t \sin \theta}{t \cos \theta}$ $m_L = \tan \theta$ L is a tangent $\Rightarrow A \neq B$ at the same point $AB^2 = 0$ $\sin^2 \theta + 12 \sin \theta \cos \theta - 28 \cos^2 \theta = 0$ $(\sin \theta - 2 \cos \theta)(\sin \theta + 14 \cos \theta) = 0$ $\sin \theta = 2 \cos \theta \quad \sin \theta = -14 \cos \theta$ $\tan \theta = 2 \quad \tan \theta = -14$ $\therefore$ angle of inclination $\tan \theta = m$ ii) let $m = 2$ $\tan \theta = \frac{2}{1}$  $\sin \theta = \frac{2}{5}, \cos \theta = \frac{1}{5}$ - due to tangent, sum of roots = $-\frac{b}{a}$ $2t_2 = \frac{\sin \theta + 6 \cos \theta}{3 \cos^2 \theta}$ $\div 2$ $t_2 = \frac{\sin \theta + 6 \cos \theta}{6 \cos^2 \theta}$ $= \frac{(\frac{2}{5}) + 6(\frac{1}{5})}{6(\frac{1}{5})^2}$ $= \frac{(\frac{8}{5})}{(\frac{6}{5})} \Rightarrow \frac{4\sqrt{5}}{3}$	① gradient m_L ① use of AB^2 to solve question ① working to answer